

Deki Electronics, which makes various types of capacitors, says factories are using robots in areas where repeatability and precision are critical. For instance, robots can weld electronic components with consistent accuracy, ensuring that every weld meets the same specifications. Hybrid models, he says, are more realistic for India's manufacturing landscape. "Robots can handle repetitive, unsafe or precision-heavy tasks while humans oversee operations, make decisions and manage complex processes," says Sharma.

Another major shift shaping the factories of the future is the emergence of 'virtual twins' or digital replicas of physical facto-

each other and to the product being manufactured," says Deepak. "That allows companies to track production, identify faults and optimise operations instantly."

FUTURE OF JOBS

This is dramatically speeding up the manufacturing pipeline. Tasks that once took weeks—such as pricing a new component—can now be completed within a day. The immediate impact is not necessarily job cuts but a dramatic boost in productivity. For instance, Gokul points out that laptops took away manual bookkeeping, librarian roles, basic record-keeping, accounting tasks, etc., but also vastly increased the number of accounts (or data) be-

and pay them all around ₹1 lakh a month. A robot to do that will cost less than one year's wages, but will be more productive, 24x7, without a break. The labour involved in this can be reskilled to do maintenance tasks or manage the dashboard to ensure the robot is doing the task correctly," Chopra says.

Engineers can focus on higher-value tasks rather than repetitive follow-ups and data gathering. Some technologists believe that manufacturing could eventually shift towards what they call a "universal factory." Gokul of Cyn-Lr envisions factories built not around specialised machines but around adaptable robots capable of performing multiple tasks.

HYBRID MODELS ARE MORE REALISTIC FOR INDIA'S MANUFACTURING LANDSCAPE, AS ROBOTS CAN HANDLE REPETITIVE TASKS

ries. Deepak N.G., Managing Director, Dassault Systèmes India, says companies are increasingly designing and testing entire factories in the virtual world before building them. "When companies plan a new plant today, they often want to build it virtually first," he says. "Every machine, every component, is modelled digitally with the same physical properties as the real equipment."

In these next-generation factories, machines do more than just perform tasks. They continuously generate data that flows through virtual platforms. Managers can track production, quality metrics and operational performance on their phones in real time.

"If I have 50 machines in a factory line, all of them should talk to

ing managed. "Accountants didn't get eliminated; they became managers of larger, more complex records. The same thing will happen on factory floors. Workers will not just be manning these machines but will also train machines to execute actions," he says.

Much like humans are training AI models to do the tasks that humans are doing, such as making videos, infographics, discovering new drugs, interpreting radiology reports and so on. Chopra of industry.ai adds, "Jobs will not disappear but change. The quality of jobs will go up."

For example, a large Noida-based manufacturer is evaluating using robots instead of humans for even tasks like lifting and shifting goods. "I employ five people

Yet, even as automation accelerates, many experts believe human workers will remain an essential part of manufacturing for years to come. Assembly operations, for instance, still require dexterity and judgement that robots struggle to replicate. Human hands can apply delicate pressure and adjust to subtle variations in materials—capabilities that remain difficult to program into machines.

As India leapfrogs into advanced manufacturing, automation will gain currency. Whether these factories run in complete darkness or under the watchful eyes of engineers, one thing is clear: the manufacturing floor is becoming as much about algorithms and data as it is about machines and metals. **BT**